Tensor isomorphism at low analytic rank

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Abstract

Let T, T' be three-tensors over a finite field $\mathbb{F}_q$ of odd characteristic, with each factor of dimension at most n, and let $a = \mathrm{ark}(T)$ be the analytic rank of the source tensor. We give a Las Vegas decision-and-search algorithm for tensor isomorphism with expected running time

$$q^{O(n+a^2)}\,\mathrm{poly}(n, \log q).$$

Neither a nor a low-rank decomposition is required as input. In particular, slice rank at most s gives $q^{O(n+s^2)}\,\mathrm{poly}(n, \log q)$. We prove the averaging, rank-cover, and marked zero-block reductions, including all three shear families in the latter. The standard algorithmic inputs are noncommutative-rank computation and odd-characteristic algorithms for ordered alternating-tuple congruence and isometry-group generation.

1 Statement and conventions

View a three-tensor as a trilinear form $T : X \times V \times Z \to \mathbb{F}_q$. Two tensors of the same factor dimensions are *isomorphic* if invertible linear substitutions in the three factors take one to the other. The factors are ordered; factor permutations are not included.

For a nontrivial additive character $\chi : \mathbb{F}_q \to \mathbb{C}^\times$, define

$$\mathrm{bias}(T) = \mathbb{E}_{x\in X,\ v\in V,\ z\in Z}\chi(T(x, v, z)), \qquad \mathrm{ark}(T) = -\log_q \mathrm{bias}(T).$$

Averaging over one factor gives the indicator that the corresponding linear functional vanishes. Thus the bias is positive, real, and independent of χ. Analytic rank is invariant under isomorphism.

Theorem 1.1. *Let q be odd and let T, T' have the same three factor dimensions, each at most n, where $n \geq 1$. There is a Las Vegas algorithm deciding whether they are isomorphic and returning an explicit isomorphism when one exists, in expected time*

$$q^{O(n+\mathrm{ark}(T)^2)}\,\mathrm{poly}(n, \log q).$$

Only the two tensors and the finite field are supplied as input.

All implicit constants are absolute. The algorithm is Las Vegas: outputs are correct, while the algebraic subroutines may have randomized running time. Polynomial dependence on numerical q, when using such a version of a standard subroutine, is absorbed by $q^{O(n)}$.

Corollary 1.2. *A source tensor of slice rank at most s admits isomorphism testing and search in $q^{O(n+s^2)}\,\mathrm{poly}(n, \log q)$, without a supplied slice decomposition. In particular, analytic rank or slice rank $O(n^{0.7})$ gives $q^{O(n^{1.4})}$ time.*

The corollary is proved in Section 5. The theorem does not assert an exponent improvement for arbitrary tensors of analytic rank $\Theta(n)$.

Standard algorithmic inputs. We use the following established algorithms.

(1) Compute the noncommutative rank of a matrix space in polynomial time over finite fields [3]. Its minimum-dimension maximum-shrunk subspace can also be computed by the canonical version of the algorithm, as recorded in [4, Proposition 7 and the following paragraph]. Section 2.2 gives an alternative $q^{O(n)}$ construction using only rank values.

(2) Over odd-order finite fields, decide congruence of two *ordered* alternating-matrix tuples and return a congruence matrix in randomized polynomial time [5, Theorem 1.7].

(3) Over odd-order finite fields, generate the full isometry group of an ordered alternating-matrix tuple in Las Vegas polynomial time [6]. Degenerate tuples are permitted.

All tuples and algebras below have polynomial dimension. We prove the additional restrictions and orbit reductions we need; no general matrix-group orbit algorithm is assumed.

The contraction argument is the averaging behind Lampert's low-rank-fiber lemma [1, Lemmas 2.1–2.2]. The rank-cover inequality appears in [2, Lemma 6.6], attributed there to Haramaty–Shpilka; we include its proof. The marked zero-block format comes from the matrix-space-equivalence reduction of [7]. We prove its completion with a quadratic cost in the boundary dimensions.

2 Matrix spaces and a useful contraction kernel

Choose bases and put $U = X^*$. Define $A_z : V \to U$ by $T(x, v, z) = x^{\mathsf{T}} A_z v$, and let $\mathcal{B} = \{A_z : z \in Z\} \leq \mathrm{Hom}(V, U)$. We may remove the kernel of $z \mapsto A_z$, the common kernel in V, and the unused output directions in U by linear algebra. Equivalently, quotient each factor by its subspace on which the trilinear form vanishes identically.

These concise dimensions are isomorphism invariants. If they differ for the two inputs, reject. Otherwise any isomorphism of the concise forms extends by arbitrary isomorphisms between their discarded directions. Concise reduction preserves bias, since discarded directions only repeat values with constant multiplicity. Handle the zero tensor directly.

Write $m = \dim \mathcal{B}$, $v = \dim V$, $u = \dim U$, so $m, u, v \leq n$ and the coefficient map is injective. Averaging gives

$$q^{-\operatorname{ark}(T)} = \mathbb{E}_{A \in \mathcal{B}} q^{-\operatorname{rank} A} = \Pr_{A \in \mathcal{B},\ x \in V}[Ax = 0]. \tag{1}$$

In the sequel $a = \operatorname{ark}(T)$.

Lemma 2.1 (A low-rank contraction kernel)**.** *Over every finite field, some $x \in V$ has*

$$\mathcal{K}_x = \{A \in \mathcal{B} : Ax = 0\}, \qquad b_x = \operatorname{codim}_{\mathcal{B}} \mathcal{K}_x$$

satisfying

$$b_x \leq a + 1, \qquad \operatorname{mrk}(\mathcal{K}_x) \leq 2\lfloor a + 2 \rfloor.$$

Here $\operatorname{mrk}(\mathcal{K}) = \max\{\operatorname{rank} A : A \in \mathcal{K}\}$.

Proof. Let $\mathcal{Z} = \{(A, x) \in \mathcal{B} \times V : Ax = 0\}$. Equation (1) gives $|\mathcal{Z}| = q^{m+v-a}$. Under the uniform distribution on $\mathcal{Z}$,

$$\mathbb{E}_{\mathcal{Z}} q^{\operatorname{rank} A} = q^a, \qquad \mathbb{E}_{\mathcal{Z}} q^{b_x} = q^a.$$

In the first sum each A occurs $q^{v - \operatorname{rank} A}$ times; in the second each x occurs $q^{m - b_x}$ times. Conditionally on x, the matrix is uniform on $\mathcal{K}_x$. Some x therefore satisfies

$$q^{b_x} + \mathbb{E}_{A \in \mathcal{K}_x} q^{\operatorname{rank} A} \leq 2q^a \leq q^{a+1}. \tag{2}$$

Thus $b_x \le a+1$ and, because the first summand is positive, $\mathbb{E}_{\mathcal{K}_x} q^{\operatorname{rank} A} < q^{a+1}$. Consequently

$$\Pr_{A\in\mathcal{K}_x}[\operatorname{rank} A > a+2] < q^{-1} \le \tfrac{1}{2}.$$

The set $G = \{A \in \mathcal{K}_x : \operatorname{rank} A \le \lfloor a+2 \rfloor\}$ contains more than half of $\mathcal{K}_x$. For every $B \in \mathcal{K}_x$, the sets G and $B+G$ intersect, so $B = A_1 - A_2$ for two members of G. Rank subadditivity proves the claim. Strict majority also holds when $q = 2$. □

2.1 Row-column covers and noncommutative rank

For $\mathcal{K} \le \operatorname{Hom}(V,U)$, let $\mathcal{K}(W) = \langle Aw : A \in \mathcal{K},\ w \in W\rangle$ and put

$$h(\mathcal{K}) = \min_{W\le V}\big(\operatorname{codim}_V W + \dim \mathcal{K}(W)\big). \tag{3}$$

This is the shrunk-subspace characterization of $\operatorname{ncrk}(\mathcal{K})$, used as its definition here. It applies to rectangular spaces. In adapted bases a minimizing W gives a zero block after deleting $\dim \mathcal{K}(W)$ rows and $\operatorname{codim} W$ columns. Conversely any such row-column cover gives a candidate in (3).

Lemma 2.2 (All-field rank-cover bound)**.** *For every matrix space over every field,* $\operatorname{ncrk}(\mathcal{K}) \le 4\operatorname{mrk}(\mathcal{K})$.

Proof. Put $r = \operatorname{mrk}(\mathcal{K})$; the case $r = 0$ is immediate. Normalize a rank-r member to $A = \operatorname{diag}(I_r, 0)$, with the evident rectangular interpretation. Let $\overline{\mathcal{K}}$ be the lower-right restriction of $\mathcal{K}$.

Fix $B \in \mathcal{K}$ and let its lower-right block have rank s. Row and column operations within that block put its pivots in an I_s block. Clear all other entries in their columns and rows using those pivot rows and columns. These operations preserve A, since its corresponding rows and columns are zero. Reordering the blocks gives

$$B = I_s \oplus D, \qquad A = 0_s \oplus A_0, \qquad \operatorname{rank} A_0 = r.$$

As $B, A+B \in \mathcal{K}$,

$$s + \operatorname{rank} D \le r, \qquad s + \operatorname{rank}(A_0 + D) \le r.$$

Thus $r \le \operatorname{rank} D + \operatorname{rank}(A_0+D) \le 2(r-s)$, whence $s \le r/2$. Therefore $\operatorname{mrk}(\overline{\mathcal{K}}) \le \lfloor r/2 \rfloor$.

Cover the first r rows and first r columns at cost $2r$. Induction covers the remaining space at cost at most $4\lfloor r/2 \rfloor$, giving $h(\mathcal{K}) \le 4r$. Everything takes place over the original field. □

2.2 A canonical minimizing pair

Put

$$\delta(W) = \dim W - \dim \mathcal{K}(W), \qquad \delta_* = \max_{W\le V} \delta(W) = v - \operatorname{ncrk}(\mathcal{K}).$$

The function δ is supermodular:

$$\delta(W_1) + \delta(W_2) \le \delta(W_1 + W_2) + \delta(W_1 \cap W_2).$$

Indeed, $\mathcal{K}(W_1 + W_2) = \mathcal{K}(W_1) + \mathcal{K}(W_2)$ and $\mathcal{K}(W_1 \cap W_2) \subseteq \mathcal{K}(W_1) \cap \mathcal{K}(W_2)$. The maximizing subspaces are therefore closed under sums and intersections. Their intersection $W_*(\mathcal{K})$ is the unique smallest maximizer. Set

$$D_*(\mathcal{K}) = \mathcal{K}(W_*(\mathcal{K})), \qquad f = \operatorname{codim}_V W_*(\mathcal{K}), \qquad d = \dim D_*(\mathcal{K}).$$

Then

$$d + f = \operatorname{ncrk}(\mathcal{K}). \tag{4}$$

Uniqueness gives covariance: if $\mathcal{K}' = P\mathcal{K}S^{-1}$, then

$$W_*(\mathcal{K}') = SW_*(\mathcal{K}), \qquad D_*(\mathcal{K}') = PD_*(\mathcal{K}).$$

Bases adapting to these subspaces can be chosen arbitrarily.

One may use the polynomial canonical-subspace algorithm cited earlier. Alternatively, the following construction requires only a noncommutative-rank value oracle and $q^{O(n)}$ overhead. Enumerate all hyperplanes $H \le V$. Such an H contains $W_*(\mathcal{K})$ if and only if

$$\dim H - \operatorname{ncrk}(\mathcal{K}|_H) = \delta_*.$$

The forward implication uses $W_* \le H$; conversely, a maximizing subspace inside H contains W_*. Intersect all hyperplanes passing the test, taking the empty intersection to be V. The result is exactly W_*. There are at most q^v hyperplanes. Using this alternative at every occurrence below changes only the constant in the $O(n)$ exponent.

Lemmas 2.1 and 2.2 show that some contraction kernel has

$$b^2 + d^2 + f^2 \le (a+1)^2 + 64(a+2)^2. \tag{5}$$

3 Completing a marked zero-block tensor

All fields in this section have odd characteristic. Partition the three variable sets as

$$(x, u), \quad (y, v), \quad (z, w),$$

with respective dimensions $(c, d), (e, f), (k, b)$. Assume that the entire xyz block is zero. The substitutions preserve the free-coordinate subspace in each factor and induce the identity on the w quotient:

$$\begin{aligned} x &\longmapsto G_x x + H_x u, & u &\longmapsto D_u u, \\ y &\longmapsto G_y y + H_y v, & v &\longmapsto D_v v, \\ z &\longmapsto G_z z + H_z w, & w &\longmapsto w, \end{aligned}$$

with all diagonal blocks invertible.

Theorem 3.1 (Quadratic boundary completion)**.** *Equivalence under these substitutions can be decided, with a substitution returned when one exists, in expected time*

$$q^{d^2+f^2}\,\mathrm{poly}(c + d + e + f + k + b, \log q).$$

All three shear blocks H_x, H_y, H_z are included.

3.1 The three coefficient layers

Enumerate $D_u \in \mathrm{GL}_d(q)$ and $D_v \in \mathrm{GL}_f(q)$, at most $q^{d^2+f^2}$ choices. Normalize each choice to $D_u = D_v = I$ by a known change of variables. Write $t = (x, y, z)^{\mathsf{T}}$ and $\alpha = (u, v, w)^{\mathsf{T}}$. The tensor polynomial has the form

$$F(t, \alpha) = t^{\mathsf{T}} Q(\alpha) t + \ell(\alpha) t + \gamma(\alpha), \tag{6}$$

where Q is symmetric and linear in α, ℓ is a row of quadratic polynomials in α, and γ is cubic in α. Divide each off-diagonal coefficient equally between the two symmetric entries of Q; this uses odd characteristic. The seven possible blocks are

layer	tensor blocks
Q	$xyw,\ xvz,\ uyz$
ℓ	$xvw,\ uyw,\ uvz$
γ	$uvw.$

These are formal polynomial identities, including over small fields.

The remaining substitutions are $t \mapsto Gt + H\alpha$, $\alpha \mapsto \alpha$, where $G = \mathrm{diag}(G_x, G_y, G_z)$ and H lies in the vector space $\mathcal{D}$ allowing only $H_{x,u}, H_{y,v}, H_{z,w}$. They give

$$Q' = G^{\mathsf{T}} Q G, \tag{7}$$

$$\ell' = (\ell + \Delta_H)G, \qquad \Delta_H = 2(H\alpha)^{\mathsf{T}} Q(\alpha), \tag{8}$$

$$\gamma' = \gamma + \ell H\alpha + (H\alpha)^{\mathsf{T}} Q(H\alpha). \tag{9}$$

First match the ordered coefficient tuples of Q under block-diagonal G. Each coefficient is supported between a single pair of free blocks. Negating its lower off-diagonal block makes it alternating; under block-diagonal substitutions the congruence equations are equivalent. Lemma A.1 reduces the block restriction to ordinary alternating-tuple congruence. Reject if it fails, otherwise normalize the inputs to a common leading tuple Q.

3.2 The linear layer is a unitary-module orbit

Form the full adjoint-pair algebra

$$\mathcal{E} = \{(r, s) : r, s \text{ block diagonal},\ r^{\mathsf{T}} Q_i = Q_i s \text{ for every } i\}.$$

Multiplication and involution are

$$(r, s)(r', s') = (rr', s's), \qquad (r, s)^* = (s, r).$$

Symmetry of the Q_i makes this an algebra with an $\mathbb{F}_q$-linear involution. Keeping both components accommodates degenerate tuples. Its unitary group is

$$\mathrm{U}(\mathcal{E}, *) = \{(G, G^{-1}) : G \text{ block diagonal},\ G^{\mathsf{T}} Q_i G = Q_i\ \forall i\}.$$

Let $\mathcal{L}$ be the space of admissible arrays ℓ and put $\mathcal{U} = \Delta(\mathcal{D})$. The first component r defines a right $\mathcal{E}$-action on $\mathcal{L}$. Transposing the adjoint identity gives $Q_i r = s^{\mathsf{T}} Q_i$, so

$$\Delta_H r = 2(H\alpha)^{\mathsf{T}} Q r = 2(sH\alpha)^{\mathsf{T}} Q = \Delta_{sH}. \tag{10}$$

Since $sH \in \mathcal{D}$, the quotient $\mathcal{M} = \mathcal{L}/\mathcal{U}$ is an explicit right $\mathcal{E}$-module, not merely a group representation.

By Lemma A.2, find a unitary element with $[\ell_1]G = [\ell_2]$, or reject. Then solve the linear system

$$\Delta_H = \ell_2 G^{-1} - \ell_1, \qquad H \in \mathcal{D}.$$

After applying this substitution both tensors have the same actual Q and ℓ.

Use the same lemma to generate the stabilizer of $[\ell]$. Lift every generator G_i by solving $\Delta_{H_i} = \ell G_i^{-1} - \ell$. Also include an $\mathbb{F}_p$-basis of $\ker \Delta$ as substitutions (I, H), where $p = \mathrm{char}\, \mathbb{F}_q$. These generate the full stabilizer of the actual pair (Q, ℓ): its projection to G is the class stabilizer, and its kernel is the additive group $\ker \Delta$.

3.3 The final layer is an additive image

For a substitution in this stabilizer, put

$$\eta(G,H) = \ell H\alpha + (H\alpha)^{\mathsf{T}} Q(H\alpha).$$

It changes only γ, by $\gamma \mapsto \gamma + \eta(G,H)$. With substitution as a right action, multiplication is

$$(G,H)(G',H') = (GG', GH' + H).$$

Every substitution fixes the anchor variables, so η is a group homomorphism to the additive coefficient space of uvw: composing two substitutions adds their anchor-only increments.

Its image is the $\mathbb{F}_p$-span of the increments of the generators just obtained. Test whether $\gamma_2 - \gamma_1$ lies in this span. If so, linear algebra gives a product of generator powers realizing it; otherwise reject this diagonal-block branch. An $\mathbb{F}_q$-span of arbitrary generator increments would not be justified.

All coefficient spaces, algebras, and tuples have polynomial dimension. If $c + e + k = 0$, compare the anchor-only tensors directly and omit the module construction. This proves Theorem 3.1.

4 The outer algorithm

Apply concise reduction to both tensors and use the source space $\mathcal{B} \leq \mathrm{Hom}(V,U)$.

Step 1. Enumerate every $x \in V$, including zero. Compute $\mathcal{K}_x$ and its canonical pair, with parameters

$$b_x = \mathrm{codim}_{\mathcal{B}} \mathcal{K}_x, \quad d_x = \dim D_*(\mathcal{K}_x), \quad f_x = \mathrm{codim}\, W_*(\mathcal{K}_x).$$

Choose x minimizing the integer $\beta(x) = b_x^2 + d_x^2 + f_x^2$. Fix that source kernel and write its parameters as b, d, f. No analytic-rank estimate is computed.

Step 2. Enumerate all target vectors y in the corresponding column factor. Form $\mathcal{K}'_y$ and its canonical pair. Discard candidates with $(b', d', f') \neq (b, d, f)$. If $x = 0$, only $y = 0$ is needed for completeness.

Step 3. Choose bases adapted to the two marked kernels and their canonical pairs. The resulting zero-block format has

$$k = \dim \mathcal{K}, \qquad c = u - d, \qquad e = v - f.$$

Free row variables annihilate $D_*(\mathcal{K})$, free column variables lie in $W_*(\mathcal{K})$, and free coefficient variables index $\mathcal{K}$. Thus the xyz block is zero.

Step 4. Enumerate the invertible correspondence between the coefficient quotients, at cost at most q^{b^2}. Normalize it to the identity and apply Theorem 3.1. Verify returned maps on the full tensors. Return a successful isomorphism, or reject if every branch fails.

Correctness. An original isomorphism induces $\mathcal{B}' = P\mathcal{B}S^{-1}$ and an invertible coefficient map. The target vector $y = Sx$ is enumerated and satisfies $\mathcal{K}'_y = P\mathcal{K}_x S^{-1}$. Covariance of the canonical pair gives the matching parameters and required parabolic row and column shapes in adapted bases. The coefficient quotient map is among those enumerated. After normalizing it, every remaining kernel automorphism and quotient-to-kernel shear is allowed by Theorem 3.1. Thus an isomorphism survives in some branch.

Conversely, a successful inner call satisfies the full tensor equations and yields an isomorphism, which extends over discarded directions. Only the kernel mark is retained in the inner call: its answer need not map the original chosen vector x to that branch's y. Discarding this additional mark enlarges the set of valid witnesses and creates no false positives. □

Running time and Theorem 1.1. Using polynomial canonical-pair computation, the source search costs $q^v \operatorname{poly}(n, \log q)$ once. Every surviving target branch has the source value of $b^2 + d^2 + f^2$, so the target search costs

$$q^{v+b^2+d^2+f^2} \operatorname{poly}(n, \log q).$$

These source and target costs add. The source search is not repeated inside every target branch. By (5),

$$b^2 + d^2 + f^2 \le (a+1)^2 + 64(a+2)^2.$$

Absorbing additive constants and polynomial factors gives $q^{O(n+a^2)} \operatorname{poly}(n, \log q)$.

Alternatively, performing the hyperplane construction of Section 2.2 throughout the source and target searches costs $q^{O(n)} \operatorname{poly}(n, \log q)$ and gives the same final bound. The randomized calls have expected polynomial running time as specified by the standard inputs. □

5 Slice rank and hidden low-rank subspaces

Slice rank $\operatorname{srk}(T)$ is the least number of summands expressing T as products of a linear form in one factor and a bilinear form in the other two; the distinguished factor may vary among summands.

Lemma 5.1. *If T vanishes on $X_0 \times V_0 \times Z_0$, whose codimensions are r_X, r_V, r_Z, then* $\operatorname{ark}(T) \le r_X + r_V + r_Z$.

Proof. For every $x \in X_0$, the map $V_0 \to Z^*$, $v \mapsto T(x, v, \cdot)$, has image in $Z_0^{\perp}$, of dimension r_Z. Its kernel occupies at least a q^{-r_Z} fraction of V_0. Therefore

$$\Pr_{x \in X,\ v \in V}[T(x, v, \cdot) = 0] \ge q^{-r_X - r_V - r_Z}.$$

The left side is the bias. □

Intersecting the kernels of the linear forms in each factor of a slice decomposition gives total codimension at most its number of summands. Hence $\operatorname{ark}(T) \le \operatorname{srk}(T)$, proving Corollary 1.2.

Corollary 5.2. *If the source coefficient space contains any subspace $\mathcal{K}$ of codimension b and maximum matrix rank at most r, then isomorphism from this source can be solved in*

$$q^{O(n+(b+r)^2)} \operatorname{poly}(n, \log q)$$

in odd characteristic, without being supplied $\mathcal{K}$. The same conclusion with r replaced by h holds if $\operatorname{ncrk}(\mathcal{K}) \le h$.

Proof. By (1),

$$q^{-\operatorname{ark}(T)} \ge q^{-b} \mathbb{E}_{A \in \mathcal{K}} q^{-\operatorname{rank} A} \ge q^{-b-r}.$$

Thus $\operatorname{ark}(T) \le b + r$. Also every matrix has rank at most the size of any row-column cover, so $\operatorname{mrk}(\mathcal{K}) \le \operatorname{ncrk}(\mathcal{K})$. □

For completeness, the concise matrix-space representation gives

$$\operatorname{srk}(T) = \min_{\mathcal{K} \le \mathcal{B}} \bigl(\operatorname{codim}_{\mathcal{B}} \mathcal{K} + \operatorname{ncrk}(\mathcal{K})\bigr).$$

A cover of $\mathcal{K}$ gives row and column slices; a coefficient complement gives the remaining slices. Conversely, group a slice decomposition by its distinguished factor, annihilate its coefficient-linear forms to obtain $\mathcal{K}$, and use the other two groups as a row-column cover. The algorithm does not perform this minimization: one-vector search already finds squared cut parameter $O(a^2 + 1)$.

6 Comparison with the algorithm in M171983

At slice rank $O(\sqrt{n})$, Corollary 1.2 gives

$$q^{O(n)}$$

time. This is a stronger parameter-dependent guarantee than the $q^{\widetilde{O}(n^{3/2})}$ guarantee of the algorithm in M171983 [7]. There is also a concrete sense in which the algorithm as displayed can still do $q^{\Theta(n^{3/2})}$ work on such inputs: its prescribed exhaustive frame loop does not adapt to slice rank. We explain both the parameter comparison and this literal-loop statement, without asserting a complexity lower bound for low-slice-rank tensor isomorphism.

6.1 Where the old exponent comes from

For an n-dimensional matrix space in $\mathrm{Mat}_n(\mathbb{F}_q)$, the manuscript chooses a target maximum matrix rank r and uses left and right restriction matrices of dimensions $s \times n$ and $n \times s$, where its individualization lemma prescribes

$$s = \lceil 3 \max\{n/r, r\} \rceil . \tag{11}$$

After choosing one source pair, it enumerates target restriction pairs. The enumeration charge is $q^{O(ns)}$. For the resulting kernel, write $h = d + f$ for its row-column boundary. The manuscript's completion algorithm costs $q^{n(d+f)} \operatorname{poly}(n, \log q)$, giving the overall parameter bound

$$q^{O(n+ns+nh)} \operatorname{poly}(n, \log q). \tag{12}$$

The initial $O(n)$ term includes the source kernel's rank check.

The manuscript uses $h = O(r \log(r + 1))$ and takes $r = \lceil \sqrt{n} \rceil$, obtaining the stated $q^{\widetilde{O}(n^{3/2})}$ bound. Even substituting the stronger $h \le 4r$ from Lemma 2.2 leaves

$$q^{O(n+n^2/r+nr)} \operatorname{poly}(n, \log q),$$

whose usual balance is $r \asymp \sqrt{n}$. The prescribed frame dimension (11) always satisfies $s \ge 3\sqrt{n}$. Merely retuning r in that prescription, or replacing only the inner solver, does not remove its $n^{3/2}$ frame-enumeration exponent. Small slice rank by itself does not change this prescription.

The two changes in the present proof play separate roles:

- The one-contraction kernel replaces the target frame search by a single-vector enumeration of size at most q^n, while giving $b, h = O(a + 1)$.
- The completion charge becomes $q^{b^2+d^2+f^2}$ instead of a charge linear in $n(d + f)$.

Using the new outer reduction with the old completion alone would give only the bound $q^{O(n+na+a^2)}$, still permitting an $n^{3/2}$ exponent at $a = \Theta(\sqrt{n})$. Together the two changes give $q^{O(n+a^2)}$.

6.2 An explicit low-slice-rank family reaching the exhaustive loop

Here is a concise negative pair for which the prescribed loop is indeed exhausted. For $j \geq 3$, define

$$F_j(x, y, z) = x_1 \sum_{i=1}^{j} y_i z_i + y_1 \sum_{i=2}^{j} x_i z_i$$

on three j-dimensional spaces. The displayed expression is a two-slice decomposition. It is concise: the coefficients of $y_1 z_1$ and $y_1 z_i$ recover the x coordinates, and the other two factors are checked in the same way.

Its zero-pair equations are

$$x_1 y_1 = 0, \qquad x_1 y_i + y_1 x_i = 0 \quad (2 \leq i \leq j).$$

The solution set is the disjoint union of the pairs with $x_1 = y_1 = 0$, the pairs with $x_1 \neq 0$ and $y = 0$, and the pairs with $y_1 \neq 0$ and $x = 0$. Hence

$$B_j := \mathrm{bias}(F_j) = q^{-2} + 2(q-1)q^{-(j+1)}. \tag{13}$$

For $q \geq 2$ and $j \geq 3$, one has $q^{-2} < B_j < q^{-1}$, so $1 < \mathrm{ark}(F_j) < 2$.

Let $k \geq 4$, $n = k^2$, and take direct sums on disjoint coordinates:

$$T = F_k^{\oplus k}, \qquad T' = F_{k-1} \oplus F_{k+1} \oplus F_k^{\oplus(k-2)}.$$

Both are concise $n \times n \times n$ tensors. Bias is multiplicative under these direct sums, and analytic rank is therefore additive. The displayed slice decompositions and $\mathrm{ark} \leq \mathrm{srk}$ give, for both tensors,

$$k < \mathrm{ark}(T), \mathrm{ark}(T') < 2k, \qquad k < \mathrm{srk}(T), \mathrm{srk}(T') \leq 2k.$$

Thus both analytic ranks and both slice ranks are $\Theta(\sqrt{n})$.

They are nonisomorphic. Put $u = q^{-2}$ and $v = 2(q-1)q^{-(k+1)}$. Then

$$B_{k-1}B_{k+1} - B_k^2 = (u + qv)(u + v/q) - (u+v)^2 = uv(q + q^{-1} - 2) > 0.$$

Consequently $\mathrm{bias}(T') > \mathrm{bias}(T)$.

For this format the manuscript's choice is $r = k$, $s = 3k$. Its displayed algorithm enumerates every

$$L' \in \mathrm{Mat}_{s \times n}(\mathbb{F}_q), \qquad R' \in \mathrm{Mat}_{n \times s}(\mathbb{F}_q)$$

and returns a negative answer only after exhausting them. There are $q^{2sn} = q^{6k^3}$ such pairs. Shape checks can discard individual candidates early but do not remove the prescribed exhaustive enumeration. Thus this literal algorithm visits $q^{\Theta(n^{3/2})}$ frames on the displayed negative pair, whereas Theorem 1.1 has the uniform bound $q^{O(n)}$ on the entire class of sources with slice rank $O(\sqrt{n})$.

This loop count concerns the algorithm as written. The example itself can be separated by its bias or exploited direct sum structure, and an implementation augmented with such tests may reject it earlier. It proves no intrinsic hardness of this family and no lower bound for other refinements of the manuscript's method. The improvement established here is a uniform guarantee for all low-analytic-rank sources, including when their useful decompositions are not supplied.

A Explicit reductions to alternating-tuple algorithms

This appendix proves the auxiliary algorithms used in Section 3, using only the two standard alternating-tuple primitives stated in Section 1.

A.1 Prescribed block-diagonal congruence

Lemma A.1. *Over odd-order finite fields, ordered alternating-tuple congruence under a prescribed block-diagonal group admits Las Vegas polynomial-time decision and search. Its full isometry group can also be generated in Las Vegas polynomial time.*

Proof. Let $V = \mathbb{F}_q^s$ with its given block decomposition and coordinate projectors Π_j. On $V \oplus V \oplus V$, put

$$\Omega_{12}(P) = \begin{pmatrix} 0 & P & 0 \\ -P^{\mathsf{T}} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \Omega_{13} = \begin{pmatrix} 0 & 0 & I \\ 0 & 0 & 0 \\ -I & 0 & 0 \end{pmatrix}.$$

Augment both tuples by the identical forms $\Omega_{12}(I), \Omega_{13}, \Omega_{12}(\Pi_j)$ for every j. Place each original form A_i in the first diagonal block, as $\operatorname{diag}(A_i, 0, 0)$.

The radicals of $\Omega_{12}(I)$ and Ω_{13} are the third and second copies. Every transporter has the shape

$$X = \begin{pmatrix} G & 0 & 0 \\ L & H & 0 \\ M & 0 & K \end{pmatrix}.$$

The identity cross forms force $H = K = G^{-\mathsf{T}}$. The $(1,2)$ projector equations give $G^{\mathsf{T}}\Pi_j G^{-\mathsf{T}} = \Pi_j$, equivalent to $G\Pi_j = \Pi_j G$. Thus G is block diagonal. The original tuple equations are precisely $G^{\mathsf{T}} A_i G = B_i$. Conversely every desired G lifts by $\operatorname{diag}(G, G^{-\mathsf{T}}, G^{-\mathsf{T}})$. The remaining shear equations are harmless, since $L = M = 0$ works.

Apply ordinary tuple congruence. In the isometry case, projection $X \mapsto G$ is a surjective group homomorphism, so projecting generators gives the full required group. Omit zero-dimensional blocks. □

A.2 Unitary-group orbits on explicit modules

An involution on an $\mathbb{F}_q$-algebra is an $\mathbb{F}_q$-linear anti-automorphism $*$ with $(a^*)^* = a$. Its unitary group is

$$\mathrm{U}(\mathcal{A}, *) = \{e \in \mathcal{A} : e^* e = e e^* = 1\}.$$

Lemma A.2. *Let $\mathcal{A}$ be a finite-dimensional unital $\mathbb{F}_q$-algebra with an $\mathbb{F}_q$-linear involution $*$, for odd q. Let M be an explicit finite-dimensional unital right $\mathcal{A}$-module. For $v, w \in M$, one can decide and find $e \in \mathrm{U}(\mathcal{A}, *)$ such that $ve = w$, and generate a vector stabilizer, in Las Vegas polynomial time in the supplied dimensions and $\log q$.*

Proof. We give an explicit reduction and retain a regular summand to justify the double-centralizer step.

A faithful hyperbolic representation. Let $W = \mathcal{A}_{\mathcal{A}} \oplus M$, with $\rho(a)$ the right-action matrix on row coordinates. On $V = W \oplus W^*$ define

$$R(a) = \operatorname{diag}(\rho(a), \rho(a^*)^{\mathsf{T}}), \qquad J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}.$$

Transpose and involution both reverse multiplication, so R is an algebra representation, faithful because it contains the regular module. Direct multiplication gives $R(a)^{\mathsf{T}} J = J R(a^*)$. Thus $R(a)$ preserves J exactly when a is unitary.

The double centralizer. Compute a basis $C_1, \ldots, C_r$ of the commutant of $R(\mathcal{A})$. Their commutant is exactly $R(\mathcal{A})$. Indeed, an operator commuting with every module endomorphism of V commutes with the projection onto its regular summand. There it commutes with every left multiplication, so is right multiplication by some a. For each $v_0 \in V$, the module map $\mathcal{A} \to V$, $x \mapsto v_0 x$, extends by zero on a complement of the regular summand to an endomorphism of V. Commuting with these maps forces the operator to send every v_0 to $v_0 a$. This proves the assertion without a semisimplicity assumption.

Encoding the unitary group. On $D = V \oplus V$, use the alternating forms

$$A_L = \operatorname{diag}(J, 0), \quad A_R = \operatorname{diag}(0, J), \quad A_\times = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}, \quad A_i = \begin{pmatrix} 0 & C_i^{\mathsf{T}} \\ -C_i & 0 \end{pmatrix}.$$

The radicals of A_L, A_R force an isometry to be $\operatorname{diag}(X, Y)$. The cross form gives $Y = X^{-\mathsf{T}}$. The A_i equations give $X^{-1} C_i X = C_i$, so $X = R(a)$ by the double centralizer. The A_L equation makes a unitary. The A_R equation is then automatic for this standard J. Thus the group is exactly

$$\{\operatorname{diag}(R(a), R(a)^{-\mathsf{T}}) : a \in \mathrm{U}(\mathcal{A}, *)\}. \tag{14}$$

Recover algebra elements from represented matrices by linear algebra.

Encoding a marked row. Embed v, w as rows in the M summand of the first copy of V inside D, writing them $\widehat{v}, \widehat{w}$. Append a three-dimensional anchor space H. Extend the old forms by zero on H, add the three elementary alternating forms on the coordinate pairs of H, and add

$$B_v = \begin{pmatrix} 0 & \widehat{v}^{\mathsf{T}} e_1^{\mathsf{T}} \\ -e_1 \widehat{v} & 0 \end{pmatrix}$$

on $D \oplus H$, with B_w on the target side.

The nondegenerate old form $A_\times$ forces H to be preserved. The common radical of the three anchor forms is D, forcing its preservation too. The anchor block is λI_3, $\lambda^2 = 1$: their individual radicals force its coordinate lines to be preserved, and its three pairwise diagonal products must all equal one. The marked equation is $\widehat{v} Z = \lambda \widehat{w}$, where Z belongs to (14). Multiplying Z by λ gives the exact row correspondence. This stays in the group, since $\lambda = \pm 1$ is a central unitary scalar.

Ordinary tuple congruence gives the transporter. For automorphisms the map $\operatorname{diag}(Z, \lambda I_3) \mapsto \lambda Z$ is a surjective group homomorphism onto the true marked-row stabilizer. Projecting generators gives the full vector stabilizer. Every construction has polynomial dimension. □

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